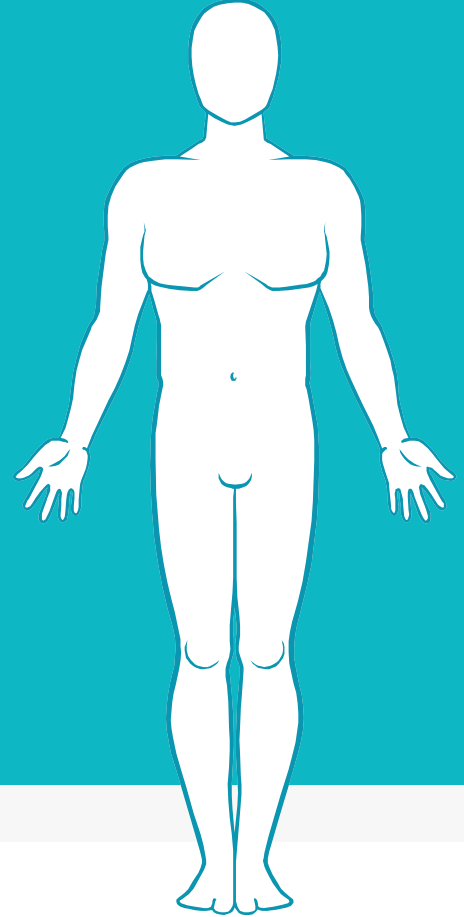


MEC E 686

Assessment and Analysis of Biomechanical Motion

Assignment 1 Guide



Data Loading

- ▶ 3 participants
- ▶ 6 trials for each participant
 - ▶ 3 trials with **right** foot landing on the force plate
 - ▶ 3 trials with **left** foot landing on the force plate
- ▶ **Format:** txt files

Data Loading

Sampling Frequency

Marker
coordinates in
global frame

Frame Number

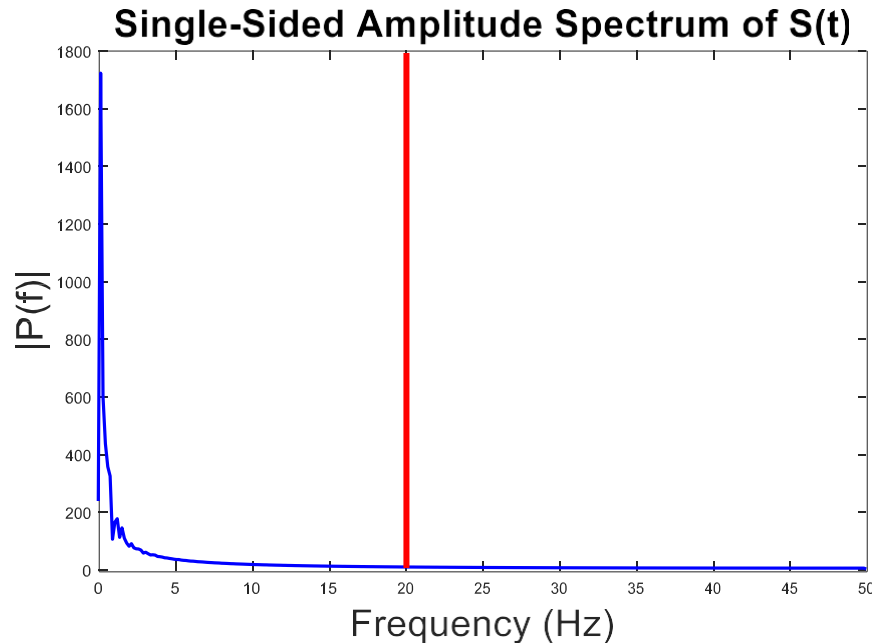
Trajectories 100																
Frame	Sub Frame	RHeel			RSMT			RToe			R1MT			R2MT		
		X	Y	Z	X	Y	Z	X	Y	Z	X	Y	Z	X	Y	Z
		mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm	mm
1	0	123.614	1550.81	11.0836	30.4634	1384.65	-4.25743	106.216	1298.11	6.31922	114.936	1340.25	15.4665	81.5615	1348.09	14.8256
2	0	123.62	1550.78	11.0585	30.4731	1384.61	-4.26381	106.227	1298.1	6.32489	114.932	1340.23	15.4669	81.5631	1348.09	14.8238
3	0	123.615	1550.79	11.0585	30.4511	1384.66	-4.2646	106.208	1298.13	6.32709	114.939	1340.22	15.4683	81.5622	1348.09	14.8277
4	0	123.616	1550.81	11.0615	30.4536	1384.65	-4.26438	106.235	1298.11	6.31629	114.938	1340.24	15.4712	81.5669	1348.1	14.8345
5	0	123.617	1550.79	11.0585	30.4788	1384.59	-4.26472	106.213	1298.12	6.30634	114.955	1340.23	15.4611	81.5332	1348.11	14.8336
6	0	123.625	1550.76	11.0364	30.4824	1384.59	-4.27312	106.223	1298.11	6.30974	114.95	1340.22	15.4589	81.5289	1348.12	14.8246
7	0	123.624	1550.77	11.0407	30.4812	1384.58	-4.27643	106.221	1298.11	6.29506	114.937	1340.24	15.4534	81.5195	1348.11	14.8405
8	0	123.625	1550.76	11.0416	30.4847	1384.58	-4.26496	106.208	1298.12	6.30406	114.949	1340.23	15.4526	81.4916	1348.15	14.8425
9	0	123.641	1550.74	11.0282	30.4851	1384.56	-4.28747	106.21	1298.11	6.30315	114.941	1340.24	15.4538	81.4834	1348.15	14.8473
10	0	123.641	1550.72	11.0251	30.506	1384.53	-4.28398	106.219	1298.08	6.29412	114.951	1340.22	15.4567	81.4888	1348.15	14.8574
11	0	123.642	1550.72	11.0164	30.4954	1384.54	-4.28181	106.205	1298.1	6.28961	114.94	1340.21	15.4653	81.4438	1348.19	14.8566
12	0	123.652	1550.69	11.0076	30.5036	1384.52	-4.2862	106.212	1298.1	6.28659	114.942	1340.21	15.4631	81.4501	1348.18	14.8764
13	0	123.653	1550.69	11.0029	30.5042	1384.5	-4.29676	106.21	1298.1	6.29216	114.94	1340.22	15.4721	81.42	1348.18	14.8726
14	0	123.654	1550.69	11.0171	30.5212	1384.48	-4.3042	106.19	1298.1	6.27921	114.933	1340.21	15.4686	81.4144	1348.19	14.8686
15	0	123.639	1550.71	11.0185	30.5182	1384.49	-4.30468	106.203	1298.11	6.29398	114.928	1340.23	15.4659	81.4201	1348.18	14.8611
16	0	123.647	1550.68	10.9935	30.5211	1384.49	-4.30825	106.185	1298.1	6.28231	114.925	1340.22	15.4592	81.4266	1348.19	14.8786
17	0	123.636	1550.71	11.011	30.522	1384.47	-4.30825	106.19	1298.1	6.2898	114.918	1340.22	15.4581	81.4283	1348.19	14.8622
18	0	123.657	1550.69	11.0052	30.5179	1384.49	-4.31209	106.178	1298.1	6.27991	114.929	1340.21	15.4534	81.4227	1348.17	14.8591
19	0	123.637	1550.7	11.0047	30.5058	1384.51	-4.30607	106.19	1298.1	6.29199	114.922	1340.22	15.4634	81.4263	1348.18	14.8602
20	0	123.64	1550.7	11.0014	30.4932	1384.49	-4.29863	106.197	1298.1	6.29646	114.924	1340.24	15.4615	81.4288	1348.19	14.8635
21	0	123.643	1550.69	11.0101	30.5154	1384.48	-4.30254	106.193	1298.11	6.29407	114.928	1340.22	15.4645	81.4317	1348.18	14.8792
22	0	123.632	1550.73	11.0191	30.5339	1384.45	-4.31583	106.187	1298.12	6.29545	114.924	1340.22	15.4601	81.4163	1348.19	14.8785
23	0	123.632	1550.7	11.0086	30.523	1384.49	-4.30162	106.196	1298.11	6.30129	114.927	1340.23	15.4612	81.4305	1348.17	14.8622
24	0	123.631	1550.71	11.0178	30.5327	1384.47	-4.30565	106.192	1298.11	6.29092	114.919	1340.21	15.4679	81.4373	1348.18	14.8716
25	0	123.632	1550.7	11.0092	30.5044	1384.47	-4.30955	106.198	1298.09	6.27708	114.934	1340.23	15.4578	81.4311	1348.18	14.8681
26	0	123.64	1550.68	11.013	30.519	1384.47	-4.31614	106.2	1298.09	6.27795	114.933	1340.23	15.4524	81.4443	1348.17	14.86
27	0	123.652	1550.69	11.0133	30.5426	1384.48	-4.30815	106.189	1298.12	6.28548	114.934	1340.22	15.4573	81.4144	1348.19	14.87
28	0	123.634	1550.7	11.0286	30.532	1384.47	-4.30474	106.206	1298.1	6.29338	114.949	1340.22	15.4639	81.4269	1348.2	14.86
29	0	123.643	1550.7	11.0225	30.5371	1384.48	-4.30533	106.21	1298.1	6.28412	114.942	1340.22	15.4436	81.426	1348.19	14.8735
30	0	123.643	1550.71	11.0253	30.5345	1384.47	-4.29341	106.212	1298.09	6.28863	114.946	1340.21	15.4531	81.4138	1348.21	14.8697
31	0	123.632	1550.7	11.0121	30.5454	1384.47	-4.29614	106.213	1298.09	6.27099	114.952	1340.21	15.4515	81.4358	1348.21	14.8813

Note:
Units are [mm]

Filtering



- ▶ Fast Fourier Transform (FFT) => Frequency content of the signal



Filtering

- ▶ Design a digital filter for attenuating the noise
- ▶ E.g., 8th-order Butterworth filter with a cut-off frequency of 20 Hz
- ▶ Normalized frequency in MATLAB: $f_n = \frac{f_c}{0.5f_s}$
- ▶ Use zero-lag (dual-pass) filter to avoid phase shift



Data Structure (one option)

- ▶ Can use data structure for motion capture data (e.g., *struct*, *cell*, etc.)
 - ▶ Easy access to each marker
 - ▶ Identification of gaps for each marker
 - ▶ Easier to debug the code
- ▶ Use column vectors instead of row vectors ($\begin{bmatrix} x \\ y \\ z \end{bmatrix}$ instead of $[x \ y \ z]$)
 - ▶ Easier when using rotation matrices

Local Coordinate System (LCS)

- ▶ Modified versions of the anatomical coordinate systems defined by Cappozzo et al. 1995

BUTTERWORTH
HEINEMANN

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Papers

Position and orientation in space of bones during movement: anatomical frame definition and determination

A Cappozzo¹, F Catani², U Della Croce¹, A Leardini²

¹Istituto di Fisiologia Umana, Università degli Studi 'La Sapienza', Roma; ²Laboratorio di Biomeccanica, Istituti Ortopedici Rizzoli, Bologna, Italy

Local Coordinate System

Pelvis

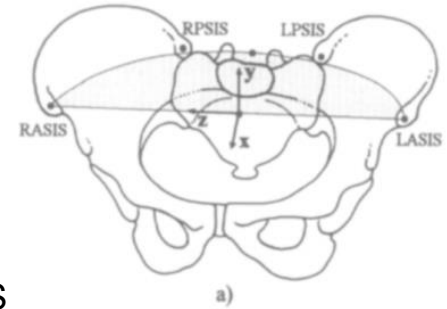
$$\vec{z} = \frac{RASIS - LASIS}{|RASIS - LASIS|}$$

$$\vec{a} = \frac{RASIS - PSIS_midpoint}{|RASIS - PSIS_midpoint|}$$

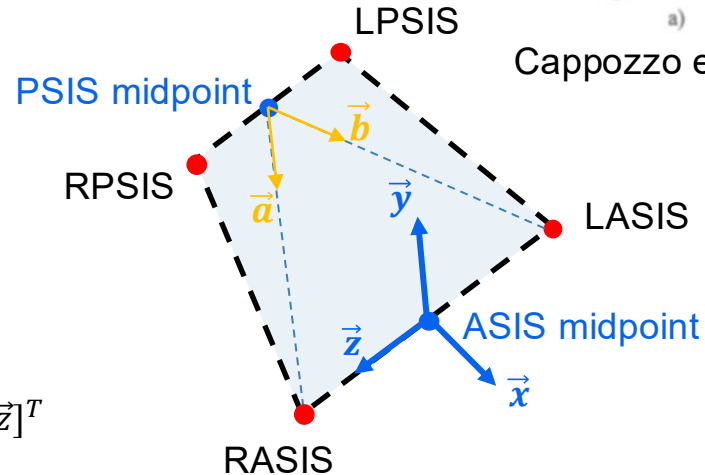
$$\vec{b} = \frac{LASIS - PSIS_midpoint}{|LASIS - PSIS_midpoint|}$$

$$\vec{y} = \frac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|} \quad \vec{x} = \frac{\vec{y} \times \vec{z}}{|\vec{y} \times \vec{z}|} \quad \Rightarrow \quad R = [\vec{x}, \vec{y}, \vec{z}]^T$$

► Origin: midpoint $P = \frac{LASIS + RASIS}{2}$



Cappozzo et al. 1995



Local Coordinate System

Thigh

$$\overrightarrow{ME2LE} = \frac{LE - ME}{|LE - ME|}$$

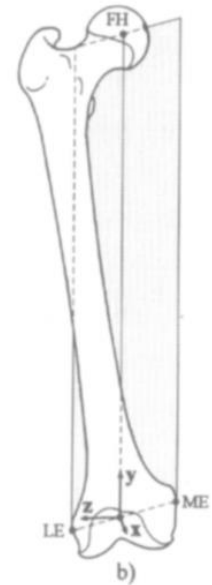
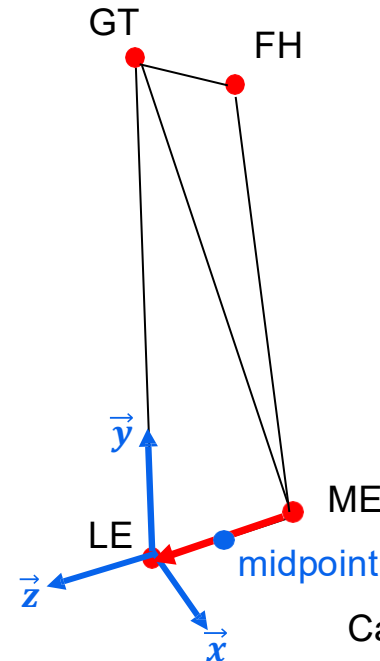
Note: for left body side
 $\overrightarrow{ME2LE} = -\overrightarrow{ME2LE}$

$$\vec{y} = \frac{GT - LE}{|GT - LE|}$$

$$\vec{x} = \frac{\vec{y} \times \overrightarrow{ME2LE}}{|\vec{y} \times \overrightarrow{ME2LE}|}$$

$$\vec{z} = \frac{\vec{x} \times \vec{y}}{|\vec{x} \times \vec{y}|} \Rightarrow R = [\vec{x}, \vec{y}, \vec{z}]^T$$

► Origin: midpoint $P = \frac{LE + ME}{2}$



Cappozzo et al. 1995

Local Coordinate System

Shank

$$\overrightarrow{MM2LM} = \frac{LM - MM}{|LM - MM|}$$

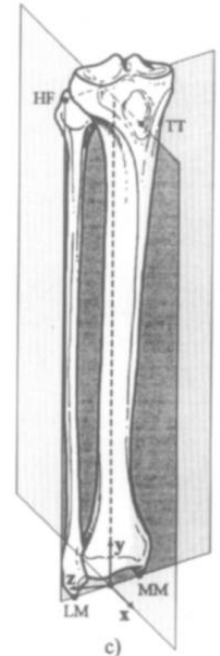
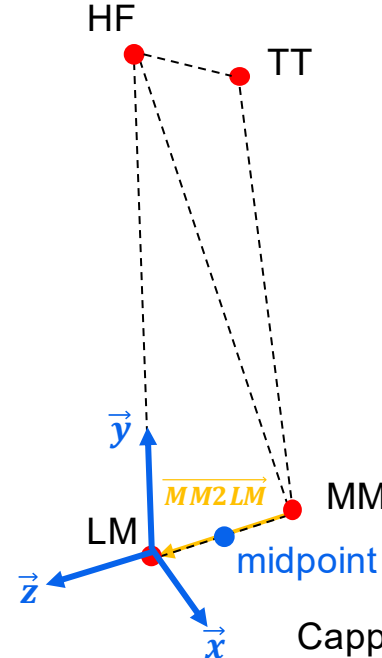
Note: for left body side
 $\overrightarrow{MM2LM} = -\overrightarrow{MM2LM}$

$$\vec{y} = \frac{HF - LM}{|HF - LM|}$$

$$\vec{x} = \frac{\vec{y} \times \overrightarrow{MM2LM}}{|\vec{y} \times \overrightarrow{MM2LM}|}$$

$$\vec{z} = \frac{\vec{x} \times \vec{y}}{|\vec{x} \times \vec{y}|} \Rightarrow R = [\vec{x}, \vec{y}, \vec{z}]^T$$

► Origin: midpoint $P = \frac{LM + MM}{2}$



Cappozzo et al. 1995



Local Coordinate System

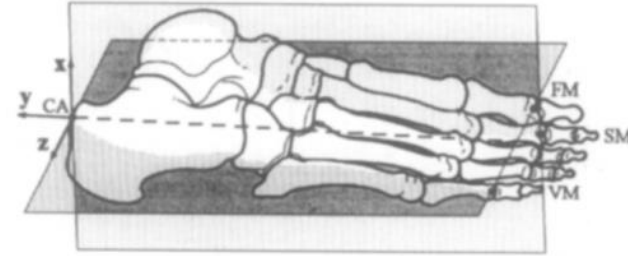
Foot

$$\vec{a} = \frac{MH1 - CA}{|MH1 - CA|} \quad \vec{b} = \frac{MH5 - CA}{|MH5 - CA|} \quad \vec{c} = \frac{MH2 - CA}{|MH2 - CA|}$$

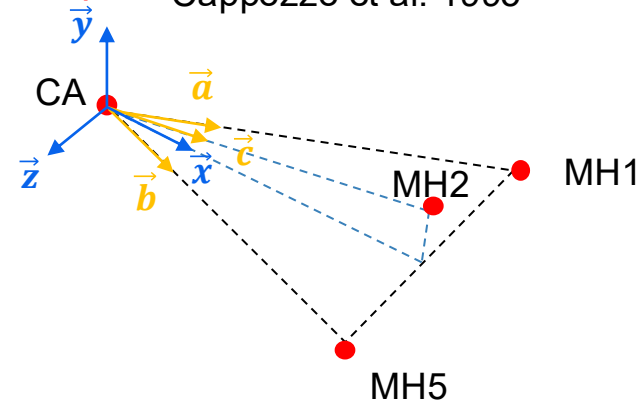
$$\overrightarrow{TP} = \frac{\vec{b} \times \vec{a}}{|\vec{b} \times \vec{a}|} \quad \overrightarrow{SP} = \frac{\vec{c} \times \overrightarrow{TP}}{|\vec{c} \times \overrightarrow{TP}|} \quad (\text{Note: for left body side } \overrightarrow{TP} = -\overrightarrow{TP})$$

$$\vec{y} = \overrightarrow{TP} \quad \vec{x} = \frac{\overrightarrow{TP} \times \overrightarrow{SP}}{|\overrightarrow{TP} \times \overrightarrow{SP}|} \quad \vec{z} = \frac{\vec{x} \times \vec{y}}{|\vec{x} \times \vec{y}|} \quad \Rightarrow R = [\vec{x}, \vec{y}, \vec{z}]^T$$

Origin: CA

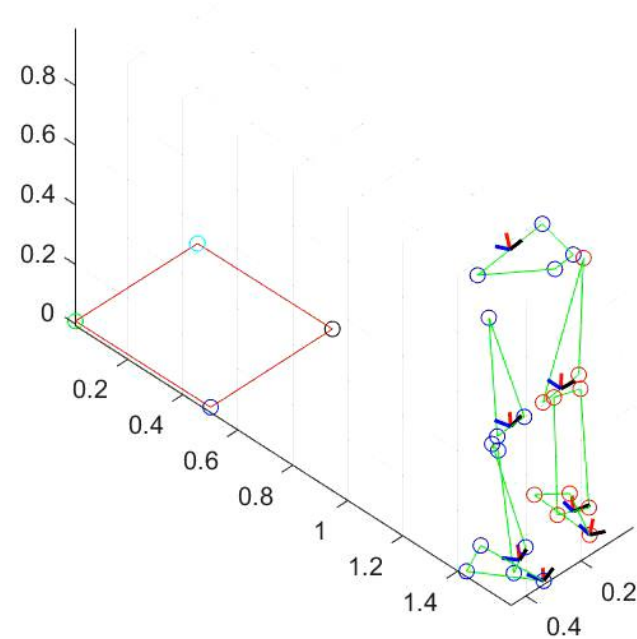


Cappozzo et al. 1995



Animation

- ▶ Better understanding the problem
- ▶ Markers and LCSs
- ▶ Identifying contra-lateral side
- ▶ Making sure everything looks OK!





Helical Angle

θ : Helical angle between proximal and distal segments

$$R = R_{proximal} \times R_{Distal}^T$$

$$\left\{ \begin{array}{l} \sin\theta = \frac{1}{2}\sqrt{(R_{23}-R_{32})^2+(R_{31}-R_{13})^2+(R_{12}-R_{21})^2} \\ \\ \text{if: } \sin\theta > \frac{\sqrt{2}}{2} \Rightarrow \theta = \cos^{-1}\left[\frac{(R_{11}+R_{22}+R_{33}-1)}{2}\right] \\ \\ \text{else: } \theta = \sin^{-1}\left[\frac{1}{2}\sqrt{(R_{23}-R_{32})^2+(R_{31}-R_{13})^2+(R_{12}-R_{21})^2}\right] \end{array} \right.$$

$$\vec{n} = \frac{\frac{1}{2}\begin{bmatrix} R_{23}-R_{32} \\ R_{31}-R_{13} \\ R_{12}-R_{21} \end{bmatrix}}{2\sin\theta}$$



Joint Coordinate System

- ▶ Grood and Suntay 1983

E. S. Grood

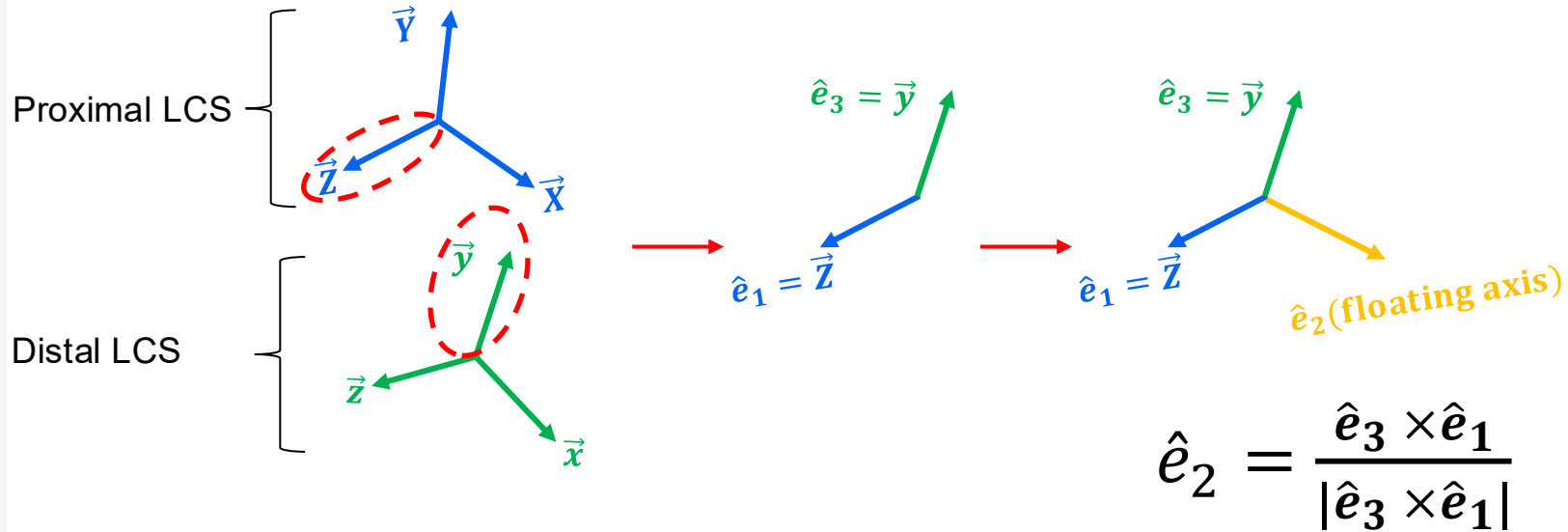
W. J. Suntay

Giannestras Biomechanics Laboratory,
Department of Orthopaedic Surgery,
University of Cincinnati Medical Center,
Cincinnati, Ohio 45267

A Joint Coordinate System for the Clinical Description of Three-Dimensional Motions: Application to the Knee¹

The experimental study of joint kinematics in three dimensions requires the description and measurement of six motion components. An important aspect of any method of description is the ease with which it is communicated to those who use the data. This paper presents a joint coordinate system that provides a simple geometric description of the three-dimensional rotational and translational motion between two rigid bodies. The coordinate system is applied to the knee and related to the commonly used clinical terms for knee joint motion. A convenient characteristic of the coordinate system shared by spatial linkages is that large joint displacements are independent of the order in which the component translations and rotations occur.

Joint Coordinate System



Joint Coordinate System (JCS) may **NOT** be an orthogonal coordinate system

Joint Coordinate System

Note: 'Prime' superscripts indicate transposition

- ▶ **Flexion/extension (F/E):** angle between proximal anterior axis (X) and floating axis ($\hat{e}_2$):

$$F/E = \text{acos}(X' * \hat{e}_2) \cdot \text{Sign} \quad \text{Sign} = \text{sign}(\text{asin}(\hat{e}_2' * Y)), \quad Y: \text{proximal vertical axis}$$

- ▶ **Adduction/abduction (A/A):** angle between proximal lateral axis (Z) and distal vertical axis (y):

$$\left\{ \begin{array}{ll} A/A = \frac{\pi}{2} - \text{acos}(Z' * y) & \text{Left body side} \\ A/A = -\frac{\pi}{2} + \text{acos}(Z' * y) & \text{Right body side} \end{array} \right.$$

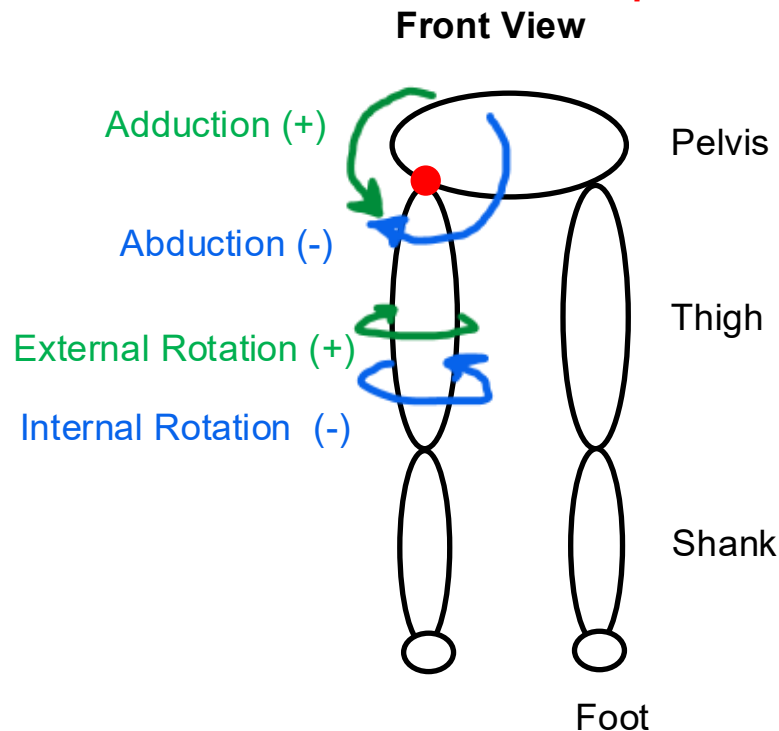
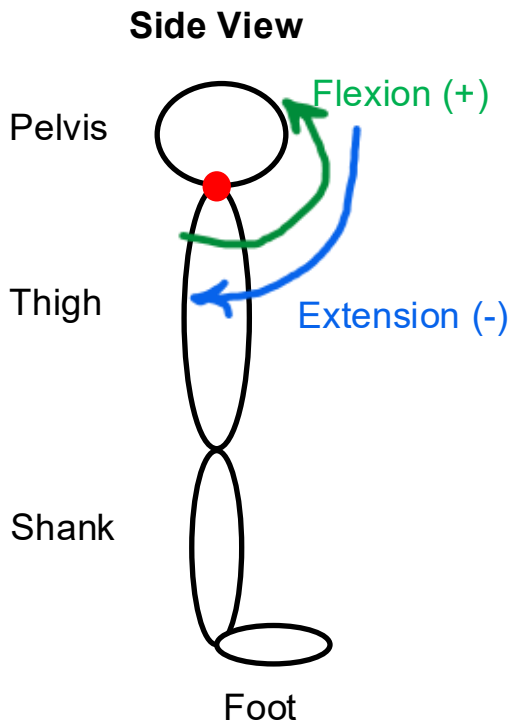
- ▶ **External-rotation/internal-rotation (ER/IR):** angle between distal anterior axis (x) and floating axis ($\hat{e}_2$):

$$\text{Sign} = \text{sign}(\text{asin}(\hat{e}_2' * z)), \quad z: \text{distal lateral axis}$$

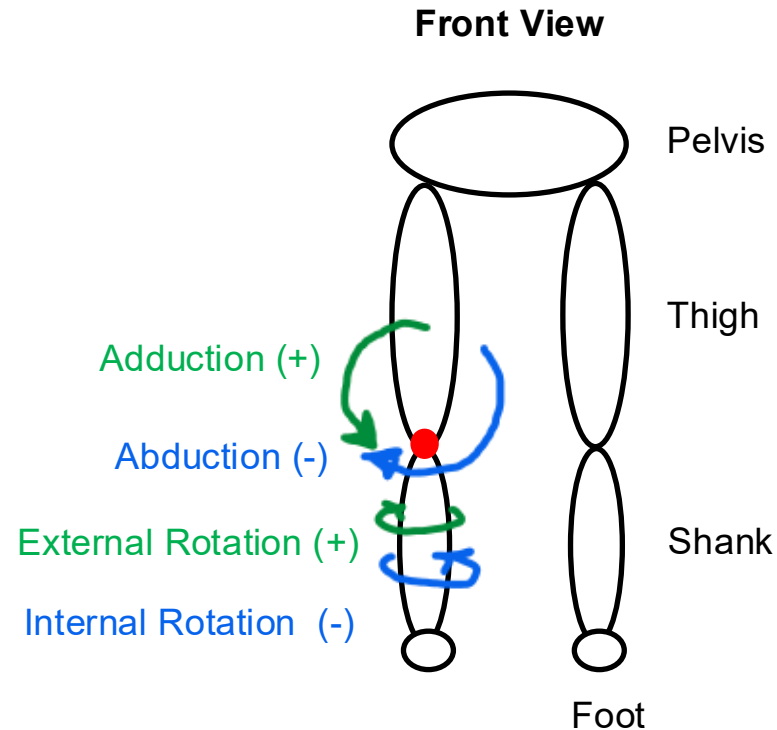
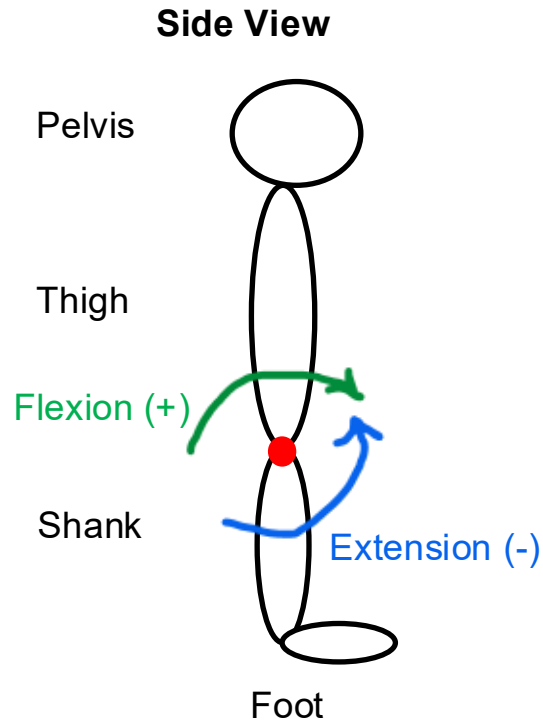
$$\left\{ \begin{array}{ll} ER/IR = -\text{acos}(x' * \hat{e}_2) \cdot \text{Sign} & \text{Left body side} \\ ER/IR = \text{acos}(x' * \hat{e}_2) \cdot \text{Sign} & \text{Right body side} \end{array} \right.$$

Sign Convention, Hip

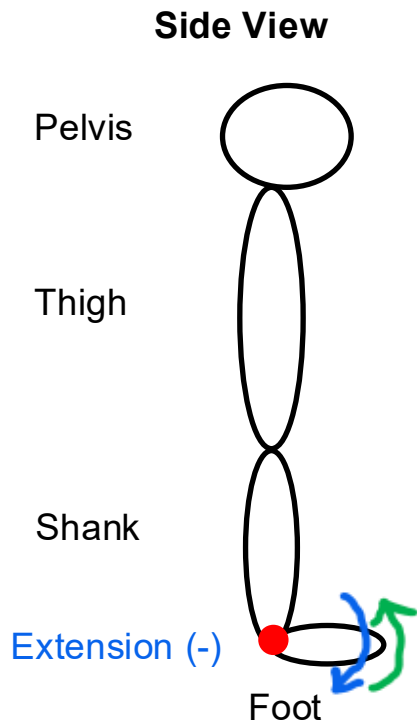
Note: For all joints, arrows indicate the direction of motion of the distal relative to the proximal segment



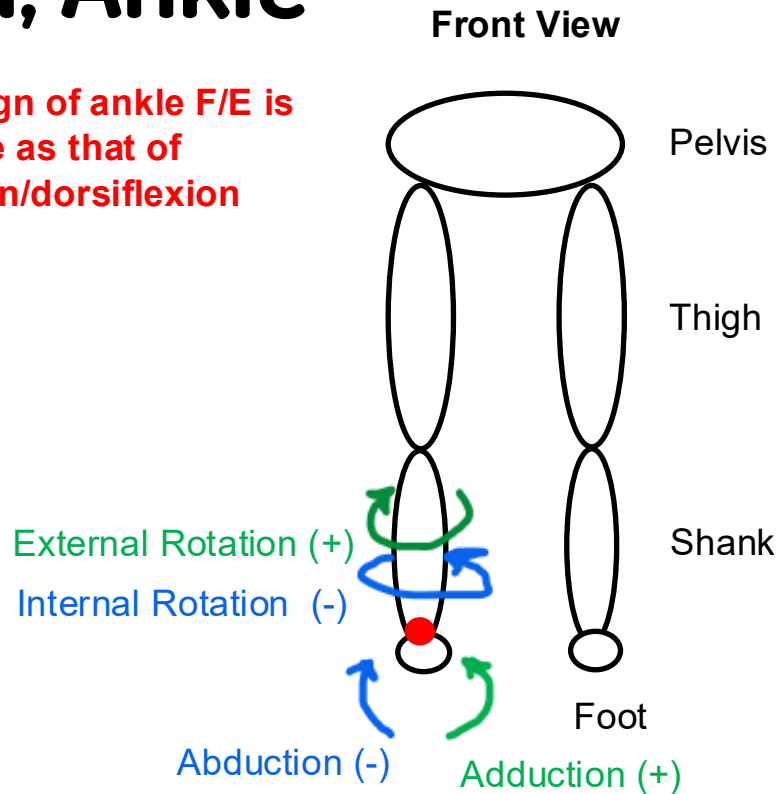
Sign Convention, Knee



Sign Convention, Ankle



Note: The sign of ankle F/E is not the same as that of plantarflexion/dorsiflexion



Gait Cycle Detection

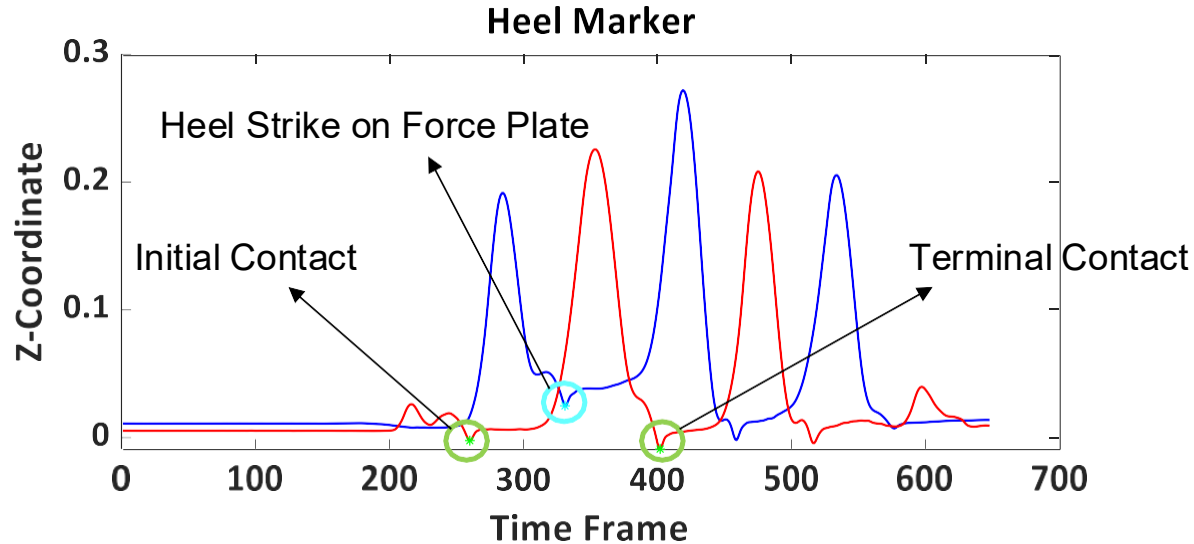
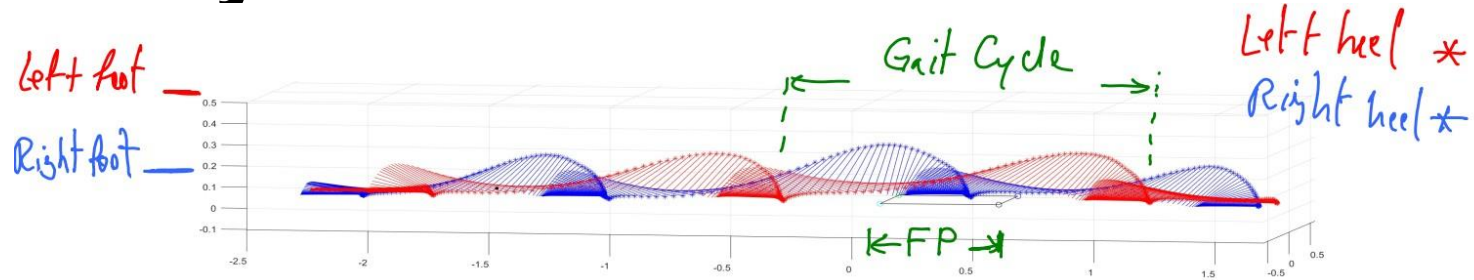
Step 1: Plot (or identify without plotting) the stance phase on the force plate

Step 2: Identify the foot that does not land on the force plate (Contra-lateral side)

Step 3: Identify the Heel-Strike instants of the contra-lateral foot that are immediately before and immediately after the force plate landing.

- ▶ Identify the minimum Z-coordinate of both heel markers
- ▶ Check which of them corresponds to force-plate landing
- ▶ Heel's Y-coordinate must fall between all four force-plate markers for this heel-strike.
- ▶ Find the initial and terminal frames for Heel-strikes of the contra-lateral side

Gait Cycle Detection



Time Normalization

- Keep the gait cycle between the Initial-Time and Terminal-time Heel-strikes of the contra-lateral side and discard the rest of the data.
- Normalize the data to 100 time-samples using Linear interpolation, Spline, or Digital filtering (resample comment in MATLAB)